Mikolaj Michal Marchelek

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Education

Georgia Institute of Technology — Atlanta, GA

Bachelor of Science in Computer Science, Minor in FinTech — *Aug. 2025 – Dec. 2028*

- GPA: 3.91/4.00 | Concentration: Intelligence, Systems & Architecture
- Relevant Coursework: Linear Algebra, Applied Combinatorics, Discrete Math, Data Structures & Algorithms, Computer Organization, Object Oriented Programing
- Associations: Snowboard Competitive Team, Polish Club @GT, Tau Kappa Epsilon Fraternity

Technical Skills

Languages: Python, Java, SQL, C, C++, Assembly
ML/AI: PyTorch, Hugging Face, FAISS, Amazon Bedrock, pandas, NumPy, scikit-learn
Infra: AWS (EKS, IAM, ECR, CloudWatch), Kubernetes, Terraform, Docker, GitHub Actions, Git, Streamlit, n8n
Spoken: Polish (native), English (fluent)

Experience

AI Software Engineer Intern — June 2026 – Sep. 2026

Mindbox Group — *Warsaw, PL*

- Architected an autonomous AI agent pipeline on Amazon EKS using n8n orchestration and Amazon Bedrock, automating Jira-to-PR code remediation with CloudWatch log analysis
- Provisioned infrastructure as code with Terraform (EKS, IAM, ECR, VPC), conducting unit tests, cutting cluster costs via node right-sizing and least-privilege IAM
- Built GitHub Actions CI/CD with Bedrock-powered automated PR review, enabling zero-touch deployment to production Kubernetes on every merge with cross-platform Docker builds.

Undergraduate Product Engineer — Sep. 2026 – Present

Virtually Integrated Project (VIP) @ Georgia Tech — *Atlanta, GA*

- AI Based Discovery and Innovation Program. Real Estate AI Product Platform Team
- Desiging system architecture for a zone-aware real estate intelligence app, wrapping property/zoning ML models in a secure API with event-driven re-scoring and explainability

Software Engineer — Sep. 2023 – May 2024

We Care for You Clinic — *Warsaw, PL*

- Built a full-stack client management application in Java (JavaFX) for a physiotherapy clinic serving 30 clients/week
- Implemented client registration, data visualization, and a PostgreSQL client database

Projects and Research

Algoverse Scholar | *AI Research Program* — May 2026 – Present

- Built a retrieval-augmented reasoning pipeline for LLMs using FAISS/E5 retrieval and token-level `<think>` boundary control on DeepSeek-R1, evaluated on AIME 2025 and GSM-8K
- Engineered constrained decoding to override base-model solving bias, raising genuine hint yield from <15% to 75%, while diagnosing failure modes like prompt dilution and hallucination
- Designed ablations on retrieval basis and hint-injection strategy, running multi-GPU experiments on shared H100/A100 infrastructure with seeded, resumable pipelines
- Preparing submission to an ARR main conference, October 2026

Machine Learning Research | *Independent Research* — May 2024

- Designed and executed a research paper under a high school advisor, comparing SVM and logistic regression for diabetes prediction
- Preprocessed data (multicollinearity, normalization) and analyzed non-linearity in larger datasets using clinical evaluation metrics in Python (Pandas, NumPy)

Limit Order Book & Matching Engine with Reinforcement Learning Agent | *Java, Python, JMH, Linux* — Oct. 2026

- Built a price-time priority matching engine in Java (red-black-tree price levels, FIFO queues, O(1) order lookup) with a stochastic market simulator and live JavaFX UI; 61 tests under CI
- Profiled hot-path performance with JMH (∼85 ns add/cancel, ∼230 ns trade), diagnosing O(n) cancel scaling and tracing ∼8% GC overhead to trade-log retention
- Trained an RL agent against the engine as a deterministic, seeded environment; deployed on Raspberry Pi 5 (ARM64 Linux) as a systemd service ingesting live crypto feeds 24/7

RAG-Based Healthcare Charge Parser | *top 10 in Hacklytics @ GT* — February 2026

- Developed an LLM-powered medical bill auditing pipeline using OpenAI GPT and pdfplumber to extract, parse, and semantically analyze itemized charges from patient PDF bills
- Built a CPT code matching engine that benchmarks patient charges against real CMS Machine-Readable Files (MRF) using scikit-learn and statistical pricing models across 255 insurance plans
- Deployed a full-stack Streamlit application with an RAG-style AI chatbot, fairness scoring algorithm, and automated dispute letter generation from structured bill intelligence